

PHYSICS

1. A steel wire of cross-sectional area $3 \times 10^{-6} \text{ m}^2$ can withstand a maximum strain of 10^{-3} . Young's modulus of steel is $2 \times 10^{11} \text{ N/m}^2$. The maximum mass the wire can hold is: (Take $g = 10 \text{ m/s}^2$)
 (1) 40 kg (2) 60 kg
 (3) 80 kg (4) 100 kg
2. A load W produces an extension of 1 mm in a thread of radius r . Now if the load is made $4W$ and radius is made $2r$ all other things remaining same, the extension will become:
 (1) 4 mm (2) 16 mm
 (3) 1 mm (4) 0.25 mm
3. A vessel has 6 g of hydrogen at pressure P and temperature 500 K. A small hole is made in it so that hydrogen leaks out. How much hydrogen leaks out if the final pressure is $\frac{P}{2}$ and temperature falls to 300 K?
 (1) 2g (2) 3g (3) 4g (4) 1g
4. Two flasks R and S of volume V_1 and V_2 contain same gas at pressure P_1 and P_2 respectively at the same temperature. Pressure of the gas when the flasks R and S are connected by a tube of negligible volume is
 (1) $\frac{P_1 V_1 + P_2 V_2}{V_1 + V_2}$ (2) $\frac{P_1 V_1 + P_2 V_2}{2(V_1 + V_2)}$
 (3) $\frac{P_1 V_2 + P_2 V_1}{V_1 + V_2}$ (4) $\frac{(P_1 + P_2)(V_1 V_2)}{(V_1 + 2V_2)}$
5. A spherical liquid drop of radius R is divided into eight equal droplets. If surface tension is T , then the work done in this process will be:
 (1) $2\pi RT$ (2) $3\pi R^2 T$
 (3) $4\pi R^2 T$ (4) $2\pi R^2 T$
6. The excess pressure inside one soap bubble is three times that inside a second soap bubble, then the ratio of their surface areas is:
 (1) 1 : 9 (2) 1 : 3 (3) 3 : 1 (4) 1 : 27
7. Water rises to a height ' h ' in capillary tube. If the length of capillary tube above the surface of water is made less than h then:
 (1) Water rises upto the top of capillary tube and stays there without overflowing.
 (2) Water rises upto a point a little below the top and stays there
 (3) Water does not rise at all
 (4) Water rises upto the top of capillary tube and then starts overflowing like fountain
8. A balloon contains 1500 m^3 of helium at 27°C and 4 atmospheric pressure. The volume of helium at -3°C temperature and 2 atmospheric pressure will be
 (1) 1500 m^3 (2) 1700 m^3
 (3) 1900 m^3 (4) 2700 m^3
9. Two capillary tubes of same diameter are put vertically one each in two liquids whose relative densities are 0.8 and 0.6 and surface tensions are 60 dyne/cm and 50 dyne/cm respectively. Ratio of heights of liquids in the two tubes $\frac{h_1}{h_2}$ is :
 (1) $\frac{10}{9}$ (2) $\frac{3}{10}$ (3) $\frac{10}{3}$ (4) $\frac{9}{10}$
10. Two spheres of the same material, but of radii R and $3R$ are allowed to fall vertically downwards through a liquid of density σ . The ratio of their terminal velocities is :
 (1) 1 : 3 (2) 1 : 6 (3) 1 : 9 (4) 4 : 1
11. A beaker is completely filled with water at 4°C . It will overflow if:
 (1) Heated above 4°C
 (2) Cooled below 4°C
 (3) Both (1) and (2)
 (4) None of the above
12. 10 g of ice at 0°C is mixed with 100g of water at 50°C . What is the resultant temperature of mixture:
 (1) 31.2°C (2) 32.8°C
 (3) 36.7°C (4) 38.2°C
13. Water is used to cool radiators of engines, because:
 (1) Of its lower density
 (2) It is easily available
 (3) It is cheap
 (4) It has high specific heat
14. It is difficult to cook rice in an open vessel by boiling it at high altitudes because of:
 (1) Low boiling point and high pressure
 (2) High boiling point and low pressure
 (3) Low boiling point and low pressure
 (4) High boiling point and high pressure

15. Two water pipes of diameters 2 cm and 4 cm are connected with the main supply line. The velocity of flow of water in the pipe of 2 cm diameter is
 (1) 4 times that in the other pipe
 (2) $\frac{1}{4}$ times that in the other pipe
 (3) 2 times that in the other pipe
 (4) $\frac{1}{2}$ times that in the other pipe
16. If V_H , V_N and V_O denote the root-mean square velocities of molecules of hydrogen, nitrogen and oxygen respectively at a given temperature, then:
 (1) $V_N > V_O > V_H$ (2) $V_H > V_N > V_O$
 (3) $V_O = V_N = V_H$ (4) $V_O > V_H > V_N$
17. One mole of ideal monoatomic gas ($\gamma = 5/3$) is mixed with one mole of diatomic gas ($\gamma = 7/5$). What is γ for the mixture? γ denotes the ratio of specific heat at constant pressure, to that at constant volume:
 (1) $3/2$ (2) $23/15$
 (3) $35/23$ (4) $4/3$

Directions Q.18 - Q.19: Choose the correct option from the following:

- (A) Both A and R are true and R is the correct explanation of A
 (B) Both A and R are true but R is NOT the correct explanation of A
 (C) A is true but R is false
 (D) A is false and R is also false

18. **Assertion (A) :** Specific heat of a gas at constant pressure is greater than its specific heat at constant volume.

Reason (R) : At constant pressure, some heat is spent in expansion of the gas.

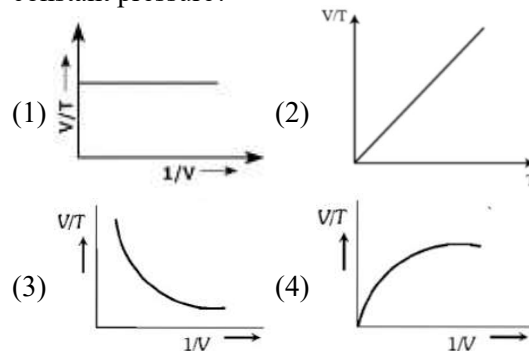
- (1) A (2) B (3) C (4) D

19. **Assertion (A) :** The total translational kinetic energy of all the molecules of a given mass of an ideal gas is 1.5 times the product of its pressure and its volume.

Reason (R) : The molecules of a gas collide with each other and the velocities of the molecules change due to collision.

- (1) A (2) B (3) C (4) D

20. Which one of the following graph is correct at constant pressure?



21. For free expansion of the ideal gas which of the following is true?

- (1) $Q = W = 0$ and $\Delta E_{\text{int}} = 0$
 (2) $Q = W > 0$ and $\Delta E_{\text{int}} = -W$
 (3) $W = 0$, $Q > 0$ and $\Delta E_{\text{int}} = Q$
 (4) $W = 0$, $Q < 0$ and $\Delta E_{\text{int}} = 0$

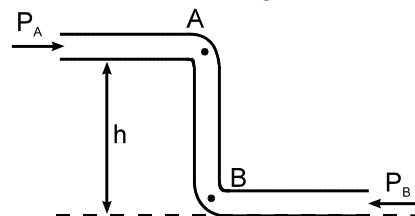
22. Mean free path of a gas molecule is

- (1) inversely proportional to number of molecules per unit volume
 (2) inversely proportional to diameter of molecule
 (3) directly proportional to the square root of the absolute temperature
 (4) directly proportional to the pressure

23. The gas law $\frac{PV}{T} = \text{constant}$ is true for:

- (1) Isothermal changes only
 (2) Adiabatic changes only
 (3) Both isothermal and adiabatic changes
 (4) Neither isothermal nor adiabatic changes

24. Figure shows an ideal fluid flowing through a uniform cross-sectional tube in the vertical tube with liquid velocities v_A and v_B and pressure P_A and P_B . Knowing that tube offers no resistance to fluid flow then which of the following is true.



- (1) $P_B > P_A$ (2) $P_B < P_A$
 (3) $P_A = P_B$ (4) none of these

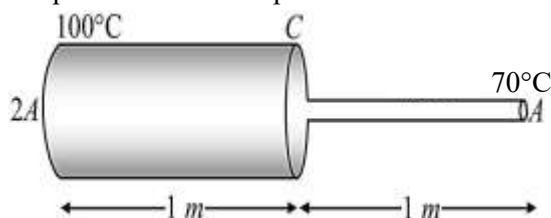
25. During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its absolute temperature. The γ of the gas is:

(1) 1.3 (2) 2
(3) 1.7 (4) 1.5

26. A perfect gas at 27°C is heated at constant pressure so as to double its volume. The increase in temperature of the gas will be

(1) 600°C (2) 327°C
(3) 300°C (4) 54°C

27. A metallic rod of length 2 m has cross-sectioned area $2A$ and A as shown in the figure. The ends are maintained at temperatures 100°C and 70°C . The temperature at middle point C is



(1) 80°C (2) 85°C
(3) 90°C (4) 95°C

28. In this question, a statement of assertion (A) is followed by a statement of reason (R). Mark the correct choice as:

Assertion: Snow is better insulator than ice.

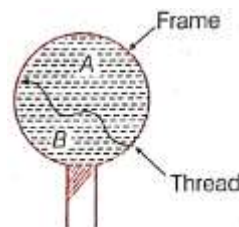
Reason: Snow contains air packet and air is good insulator of heat.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
(2) If both assertion and reason are true but reason is not the correct explanation of assertion.
(3) If assertion is true but reason is false.
(4) If assertion is false but reason is true.

29. A lead bullet, of initial temperature 27°C and speed v km/h penetrates into a solid object and melts. If 50 % of the kinetic energy is used to heat it, the value of v in kmh^{-1} is (for lead melting point = 600°C , latent heat of fusion = $2.5 \times 10^4 \text{ J/kg}$, specific heat = $125 \text{ Jkg}^{-1} \text{ K}^{-1}$)

(1) 3600 (2) 1800
(3) 1200 (4) 1000

30. A thread is tied slightly loose to a wire frame as shown in figure and the frame is dipped into a soap solution and taken out. The frame is completely covered with the film.



When A is pricked

- (1) Thread will become concave on seeing from side A
(2) Thread will become concave on seeing from side B
(3) Thread will become straight
(4) Thread will remain as it is

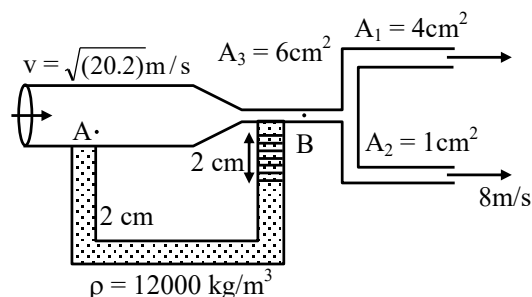
31. In this question, a statement of assertion (A) is followed by a statement of reason (R). Mark the correct choice as:

Assertion: A large soap bubble expands while a small bubble shrinks, when they are connected to each other by a capillary tube.

Reason: The excess pressure inside bubble (or drop) is inversely proportional to the radius.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
(2) If both assertion and reason are true but reason is not the correct explanation of assertion.
(3) If assertion is true but reason is false.
(4) If assertion is false but reason is true.

32. Calculate the velocity with which the liquid gushes out of the 4 cm^2 outlet, if the liquid flowing in the tube is water and liquid in U tube has a specific gravity 12. Velocity of liquid at point A is $\sqrt{20.2} \text{ m/s}$



- (1) 2.5 m/s (2) 5.5 m/s
(3) 8 m/s (4) 10 m/s

33. The speeds of air-flow on the upper and lower surfaces of a wing of an aeroplane are v_1 and v_2 respectively. If A is the cross-sectional area of the wing and ρ is the density of air, then the upward lift is

(1) $\frac{1}{2}\rho A(v_1 - v_2)$ (2) $\frac{1}{2}\rho A(v_1 + v_2)$
 (3) $\frac{1}{2}\rho A(v_1^2 - v_2^2)$ (4) $\frac{1}{2}\rho A(v_1^2 + v_2^2)$

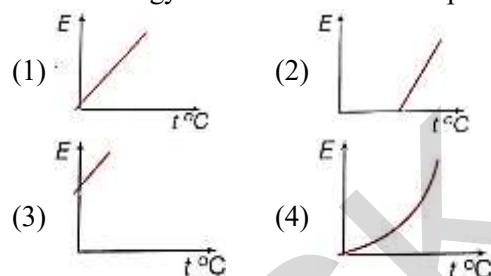
34. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Clothes containing oil or grease stains cannot be cleaned by water wash.

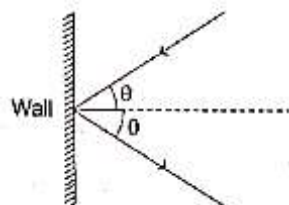
Reason (R): Because the angle of contact between the oil/grease and water is obtuse.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (3) (A) is true but (R) is false.
 (4) (A) is false but (R) is true.

35. The graph which represents the vibration of mean kinetic energy of molecules with temperature $t^\circ\text{C}$ is

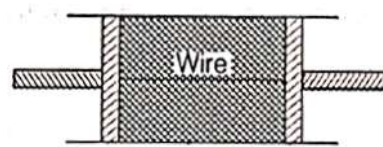


36. The mass of hydrogen molecule is 3.32×10^{-27} kg. If 10^{23} hydrogen molecules strike per second at 2 cm^2 area of a rigid wall at an angle of 45° from the normal and rebound back with a speed of 1000 ms^{-1} , then the pressure exerted on the wall is



(1) $2.34 \times 10^3\text{ Pa}$ (2) $0.23 \times 10^6\text{ Pa}$
 (3) $0.23 \times 10^3\text{ Pa}$ (4) $23.4 \times 10^3\text{ Pa}$

37. A cylindrical tube of uniform cross-sectional area A is fitted with two air tight frictionless pistons.



The pistons are connected to each other by a metallic wire. Initially, the pressure of the gas is p_0 and temperature is T_0 , atmospheric pressure is also p_0 .

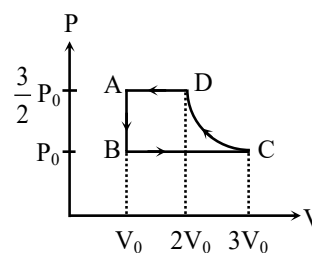
Now, the temperature of the gas is increased to $2T_0$, the tension in the wire will be

(1) $2p_0A$ (2) p_0A
 (3) $\frac{p_0A}{2}$ (4) $4p_0A$

38. Find the amount of work done to increase the temperature of one mole of an ideal gas by 30°C if it is expanding under the condition $V \propto T^{2/3}$ -

(1) 166.2 J (2) 136.2 J
 (3) 126.2 J (4) None of these

39. For the following graph match the column I with column II.



Column-I	Column-II
(a) AB	(p) $\Delta U = 0$
(b) BC	(q) $\Delta W < 0$
(c) CD	(r) $\Delta W = 0$
(d) DA	(s) $\Delta W > 0$

- (1) $a \rightarrow s, b \rightarrow p, c \rightarrow q, d \rightarrow r$
 (2) $a \rightarrow r, b \rightarrow s, c \rightarrow p, d \rightarrow q$
 (3) $a \rightarrow r, b \rightarrow q, c \rightarrow p, d \rightarrow s$
 (4) $a \rightarrow q, b \rightarrow p, c \rightarrow r, d \rightarrow s$

40. Assertion: The temperature of a gas rises during an adiabatic compression, although no heat is given from outside.

Reason: During adiabatic compression pressure of gas decreases.

- (1) If both Assertion and Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion and Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion and Reason are False.

41. A diatomic gas ($\gamma = 1.4$) does 2000 J of work when it is expanded isobarically. Find the heat given to the gas in this process (in kJ).

- (1) 5 (2) 7 (3) 9 (4) 11

42. A mono atomic gas at a pressure P, having a volume V expands isothermally to a volume 2 V and then adiabatically to a volume 16 V. What is the final pressure of the gas? (take, $\gamma = \frac{5}{3}$)

- (1) 64 P (2) 32 P
(3) $\frac{P}{64}$ (4) 16 P

43. Match the column-I and column-II and select the correct choice out of the four options.

Column-I		Column-II	
(A)	Zeroth law of thermodynamics	(I)	Give us the concept of temperature
(B)	First law of thermodynamics	(II)	Defines internal energy
(C)	Second law of thermodynamics	(III)	Change in internal energy is zero
(D)	In a complete cycle of changes	(IV)	The efficiency of on engine is always less than unity

- (1) A-(III); B-(II); C-(I, IV); D-(IV)
- (2) A-(I); B-(III); C-(IV); D-(II)
- (3) A-(II); B-(I); C-(III); D-(IV)
- (4) A-(I); B-(II); C-(IV); D-(III)

44. A perfect gas goes from state A to state B by absorbing 8×10^5 J of heat and doing 6.5×10^5 J of external work. It is now transferred between the same two states in another process in which it absorbs 10^5 J of heat. In the second process,

- (1) work done on the gas is 10^5 J
- (2) work done on the gas is 0.5×10^5 J
- (3) work done by the gas is 10^5 J
- (4) work done by the gas is 0.5×10^5 J

45. Assertion (A): Efficiency of a reversible heat engine will be highest at -273°C temperature of cold reservoir.

Reason (R): The efficiency of Carnot's engine depends not only on temperature of cold reservoir but it depends on the temperature of hot reservoir too and is given as $\eta = \left(1 - \frac{T_2}{T_1}\right)$.

In the light of the above statements, choose the correct answer from the options given below.

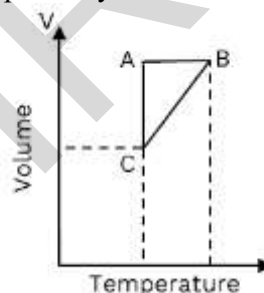
- (1) (A) is true but (R) is false.
- (2) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (3) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
- (4) (A) is false but (R) is true.

CHEMISTRY

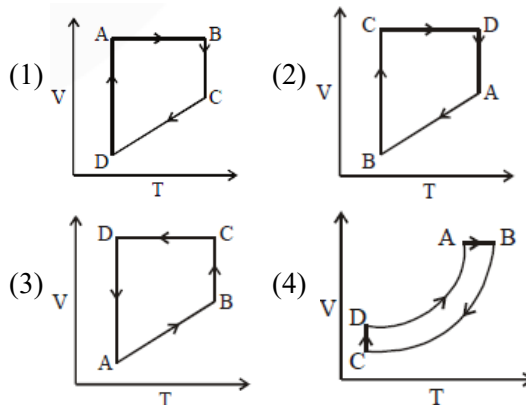
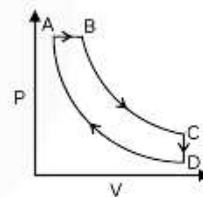
46. The reducing character of hydrides of group 14 elements is :
 (1) Maximum for CH_4 and minimum for PbH_4
 (2) Maximum for CH_4 and minimum for SnH_4
 (3) Maximum for PbH_4 and minimum for SiH_4
 (4) Maximum for PbH_4 and minimum for CH_4
47. Ga^{1+} acts as a reducing agent because—
 (1) Ga^{3+} state is less stable than Ga^{+1}
 (2) Ga^{3+} state is more stable than Ga^{+1}
 (3) Ga^{3+} Covalent to Ga^{+1} reducing
 (4) None of the above
48. Amorphous boron on burning in air forms :
 (1) $\text{B}(\text{OH})_3$ (2) Mixture of B_2O_3 and BN
 (3) Only B_2O_3 (4) Only BN
49. Which of the following statements is correct ?
 (1) Aluminium carbide as well as beryllium carbide produce methane gas on treatment with water
 (2) On reacting with water, calcium carbide (CaC_2) produces acetylene while magnesium carbide (Mg_2C_3) gives propyne
 (3) Both of these
 (4) None of these
50. Which compound does not exist?
 (1) TlCl_3 (2) TlI_3 (3) BiF_3 (4) PbI_4
51. Borax $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$ is actually :-
 (1) $\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$
 (2) $\text{Na}_2[\text{B}_4\text{O}_4(\text{OH})_6] \cdot 7\text{H}_2\text{O}$
 (3) $\text{Na}_2[\text{B}_4\text{O}_3(\text{OH})_8] \cdot 6\text{H}_2\text{O}$
 (4) $\text{Na}_2[\text{B}_4\text{O}_2(\text{OH})_{10}] \cdot 5\text{H}_2\text{O}$
52. Which one of the following mixed sulphates is not an alum?
 (1) $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
 (2) $\text{K}_2\text{SO}_4 \cdot \text{Cr}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
 (3) $\text{Na}_2\text{SO}_4 \cdot \text{Fe}(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
 (4) $\text{CuSO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
53. The type of hybridization of boron in diborane is:
 (1) sp (2) sp^2 (3) sp^3 (4) sp^3d^2
54. Which of the following species are expected to be planar?
 (a) NH_3 (b) NH_3^{2+} (c) CH_3^+ (d) PCl_3
 The correct answer is:-
 (1) b and c (2) c and d
 (3) b and d (4) a and d
55. Among the following species identify the isostructural pairs:-
 NF_3 , NO_3^- , BF_3 , H_3O^+ , HN_3
 (1) $[\text{NF}_3, \text{NO}_3^-]$ and $[\text{BF}_3, \text{H}_3\text{O}^+]$
 (2) $[\text{NF}_3, \text{HN}_3]$ and $[\text{NO}_3^-, \text{BF}_3]$
 (3) $[\text{NF}_3, \text{H}_3\text{O}^+]$ and $[\text{NO}_3^-, \text{BF}_3]$
 (4) $[\text{NF}_3, \text{H}_3\text{O}^+]$ and $[\text{HN}_3, \text{BF}_3]$
56. Which order of decreasing bond angle is correct?
 (1) $\text{CCl}_4 > \text{BF}_3 > \text{NO}_2^+$
 (2) $\text{NH}_3 > \text{NCl}_3 > \text{NBr}_3$
 (3) $\text{Br}_2\text{O} > \text{Cl}_2\text{O} > \text{OF}_2$
 (4) $\text{PCl}_3 > \text{PBr}_3 > \text{PI}_3$
57. In a closed insulated container, a liquid is stirred with a paddle to increase its temperature. In this process, Which of the following is true
 (1) $\Delta E = w = q = 0$
 (2) $\Delta E \neq 0, q = w = 0$
 (3) $\Delta E = w \neq 0, q = 0$
 (4) $\Delta E = q \neq 0, w = 0$
58. Out of boiling point (I), entropy (II), pH (III) and emf of a cell (IV), intensive properties are :
 (1) I, III, IV (2) I, II
 (3) I, II, III (4) All of these
59. A mixture of 2 moles of carbon monoxide and one mole of oxygen in a closed vessel is ignited to get carbon dioxide. If ΔH is the enthalpy change and ΔE is the change in internal energy then :-
 (1) $\Delta H > \Delta E$ (2) $\Delta H < \Delta E$
 (3) $\Delta H = \Delta E$ (4) Not definite
60. A system has internal energy equal to E_1 , 450 J of heat is taken out of it and 600 J of work is done in it. The final energy of the system will be
 (1) $(E_1 + 150)$ (2) $(E_1 + 1050)$
 (3) $(E_1 - 150)$ (4) None of these
61. For the reaction $\text{A}(\text{s}) \longrightarrow \text{B}(\text{s}) + \text{C}(\text{g})$ the value of $\Delta H = 30.56 \text{ kJ mol}^{-1}$ and $\Delta S = 66 \text{ JK}^{-1} \text{ mol}^{-1}$. The temperature at which the free energy change for the reaction will be zero is :-
 (1) 373 K (2) 413 K
 (3) 463 K (4) 493 K
62. The enthalpy of combustion of cyclohexane, cyclohexene and H_2 are respectively - 3920, - 3800 and - 241 KJmol^{-1} . The heat of hydrogenation of Cyclohexene is :-
 (1) - 121 KJ mol^{-1} (2) 121 KJ mol^{-1}
 (3) - 242 KJ mol^{-1} (4) 242 KJ mol^{-1}

63. In which of the following process work is independent of path.
 (1) Isothermal (2) Isochoric
 (3) Adiabatic (4) Isobaric
64. Which of the following is true for the reaction $\text{H}_2\text{O}(\ell) \rightleftharpoons \text{H}_2\text{O}(\text{g})$ at 100°C and 1 atm
 (1) $\Delta S = 0$ (2) $\Delta H = 0$
 (3) $\Delta H = \Delta E$ (4) $\Delta H = T\Delta S$
65. If 900 J/g of heat is exchanged at boiling point of water, then what is increase in entropy?
 (1) $43.4 \text{ JK}^{-1} \text{ mole}^{-1}$ (2) $87.2 \text{ JK}^{-1} \text{ mole}^{-1}$
 (3) $900 \text{ JK}^{-1} \text{ mole}^{-1}$ (4) Zero
66. For a reaction $2\text{X}(\text{s}) + 2\text{Y}(\text{s}) \rightarrow 2\text{C}(\text{l}) + \text{D}(\text{g})$. The q_p at 27°C is $-28 \text{ kcal. mol}^{-1}$. The q_v is ----- kcal. mol^{-1} :-
 (1) -27.4 (2) $+27.4$ (3) -28.6 (4) 28.6
67. In which reaction ΔS is positive?
 (1) $\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{O}(\text{s})$
 (2) $3\text{O}_2(\text{g}) \rightarrow 2\text{O}_3(\text{g})$
 (3) $\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{O}(\text{g})$
 (4) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$
68. The enthalpy of vapourisation for water is $186.5 \text{ kJ mol}^{-1}$, the entropy of its vapourisation will be-
 (1) $0.5 \text{ kJ K}^{-1} \text{ mole}^{-1}$ (2) $1.0 \text{ kJ K}^{-1} \text{ mole}^{-1}$
 (3) $1.5 \text{ kJ K}^{-1} \text{ mole}^{-1}$ (4) $2.0 \text{ kJ K}^{-1} \text{ mole}^{-1}$
69. A gas is allowed to expand under reversible adiabatic conditions what is zero for such a process:-
 (1) $\Delta G = 0$ (2) $\Delta T = 0$
 (3) $\Delta S = 0$ (4) None of these
70. The spontaneous nature of a reaction is impossible if :
 (1) ΔH is +ve, ΔS is also +ve
 (2) ΔH is -ve; ΔS is also -ve
 (3) ΔH is -ve ; ΔS is +ve
 (4) ΔH is +ve; ΔS is -ve
71. The values of ΔH and ΔS for the reaction, $\text{C}(\text{graphite}) + \text{CO}_2(\text{g}) \rightarrow 2\text{CO}(\text{g})$ are 170 kJ and 170 JK^{-1} , respectively. This reaction will be spontaneous at :
 (1) 510 K (2) 710 K
 (3) 910 K (4) 1100 K
72. In which of following there is decrease in entropy?
 (1) When temperature is raised from 3K to 150K
 (2) When NaHCO_3 changes into $\text{Na}_2\text{CO}_3(\text{s})$ and $\text{CO}_2(\text{g})$
 (3) $\text{H}_2(\text{g}) \rightarrow 2\text{H}(\text{g})$
 (4) Liquid crystallises into a solid

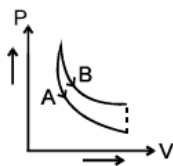
73. For endothermic reaction when change in entropy is negative, then reaction is :
 (1) not possible at any temperature
 (2) possible at low temperature
 (3) possible at all temperature
 (4) possible at high temperature
74. In which one of the following sets, all the properties belong to same category (all extensive or all intensive)?
 (1) Mass, volume, pressure
 (2) Temperature, pressure, volume
 (3) Heat capacity, density, entropy
 (4) Enthalpy, internal energy, volume.
75. Five moles of a gas is put through a series of changes as shown graphically in a cyclic process the A B, B C and C A respectively are



- (1) Isochoric, Isobaric, Isothermal
 (2) Isobaric, Isochoric, Isothermal
 (3) Isothermal, Isobaric, Isochoric
 (4) Isochoric, Isothermal, Isobaric
76. A cyclic process ABCD is shown in PV diagram for an ideal gas. which of the following diagram represents the same process?



77. In figure, A and B are two adiabatic curves for two different gases. Then A and B corresponds to:



- (1) Ar and He respectively
 (2) He and H₂ respectively
 (3) O₂ and H₂ respectively
 (4) H₂ and He respectively
78. One mole of an ideal gas $\left(C_{v,m} = \frac{5}{2}R\right)$ at 300 K and 5 atm is expanded adiabatically to a final pressure of 2 atm against a constant pressure of 2 atm. Final temperature of the gas is:
 (1) 270 K (2) 273 K
 (3) 248.5 K (4) 200 K
79. There are two samples of same gas initially under similar initial state. Gases of both the samples are expanded. Ist sample using reversible isothermal process and IInd sample using reversible adiabatic process till final pressures of both the samples becomes half of initial pressure, then
 (1) Final volume of Ist sample < final volume of IInd sample
 (2) Final volume of IInd sample < final volume of Ist sample
 (3) Final volumes will be equal
 (4) Information is insufficient
80. **Statement-1:** There is no change in enthalpy of an ideal gas during compression at constant temperature.
Statement-2: Enthalpy of an ideal gas is a function of temperature and pressure.
 (1) Statement-1 is true, statement-2 is true and statement-2 is the correct explanation for statement-1.
 (2) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (3) Statement-1 is true, statement-2 is false.
 (4) Statement-1 is false, statement-2 is true.

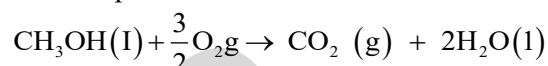
81. Suppose that the volume of a certain ideal gas is to be doubled by one of the following processes:

(A) isothermal expansion
 (B) adiabatic expansion
 (C) free expansion in insulated condition
 (D) expansion at constant pressure.

If E₁, E₂, E₃ and E₄ respectively are the changes in average kinetic energy of the molecules for the above four processes, then-

- (1) E₂ = E₃ (2) E₁ > E₃
 (3) E₁ > E₄ (4) E₄ > E₃

82. For complete combustion of methanol

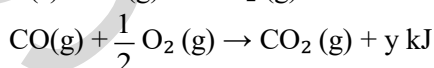
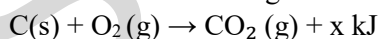


the amount of heat produced measured by bomb calorimeter is 726 kJ mol⁻¹ at 27°C. The enthalpy of combustion for the reaction is x kJ mol⁻¹, where x is (Nearest integer)

(Given: R 8.3 JK⁻¹ mol⁻¹)

- (1) 727 (2) 756 (3) 827 (4) 856

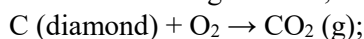
83. Consider the following reactions:



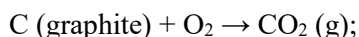
The heat of formation of CO (g) is:

- (1) (x + y) kJ/mol (2) (x - y) kJ/mol
 (3) (y - x) kJ/mol (4) None of these

84. For the following reaction,



$$\Delta H = -94.3 \text{ kcal}$$

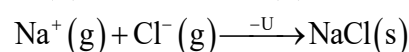
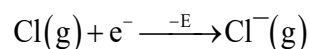
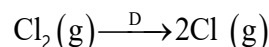
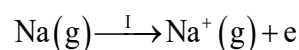


$$\Delta H = -97.6 \text{ kcal}$$

The heat required to change 1 g of C (diamond) → C (graphite) is:

- (1) 1.59 kcal (2) 0.1375 kcal
 (3) 0.55 kcal (4) 0.275 kcal

85. $\text{Na}(s) \xrightarrow{S} \text{Na}(g)$



Then ΔH_f of NaCl is :-

- (1) S + I + D - E - U
 (2) S + I + D / 2 + E + U
 (3) S + I + D / 2 - E - U
 (4) S + I + D / 2 - E + U

86. Match the following:

Column-A

- (i) Adiabatic process
- (ii) Isolated system
- (iii) Isothermal change
- (iv) Path function

Column-B

- (a) No transfer of heat
- (b) No exchange of energy and matter
- (c) Constant temperature
- (d) Heat

- | | (i) | (ii) | (iii) | (iv) |
|-----|-----|------|-------|------|
| (1) | a | b | c | d |
| (2) | b | a | c | d |
| (3) | d | c | b | a |
| (4) | a | d | b | c |

87. Match the following parameters with description for spontaneity -

Δ (Parameters)

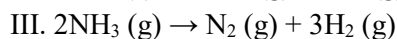
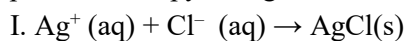
- | | $\Delta_r H^\ominus$ | $\Delta_r S^\ominus$ | $\Delta_r G^\ominus$ | |
|-------|----------------------|----------------------|----------------------|-----------|
| (i) | + | - | + | |
| (ii) | - | - | + | at high T |
| (iii) | - | + | - | |

Description

- (a) Non-spontaneous at high temperature.
- (b) Spontaneous at all temperatures
- (c) Non-spontaneous at all temperatures

- | | (i) | (ii) | (iii) |
|-----|-----|------|-------|
| (1) | c | a | b |
| (2) | b | c | a |
| (3) | a | b | c |
| (4) | c | b | a |

88. Predict which of the following reaction (s) has a positive entropy change?



- (1) I and II
- (2) III
- (3) II and III
- (4) II

89. $\text{H}_2 (\text{g}) + \frac{1}{2} \text{O}_2 (\text{g}) \rightarrow \text{H}_2\text{O} (\text{l})$

B.E. (H - H) = x_1 ; B.E. (O = O) = x_2 ;

B.E. (O - H) = x_3

Heat of vaporisation of water = x_4 then ΔH_f (heat of formation of liquid water) is:

- (1) $x_1 + \frac{x_2}{2} - x_3 + x_4$
- (2) $2x_3 - x_1 - \frac{x_2}{2} - x_4$
- (3) $x_1 + \frac{x_2}{2} - 2x_3 - x_4$
- (4) $x_1 + \frac{x_2}{2} - 2x_3 + x_4$

90. **Statement I:** The expansion of gas into an evacuated space takes place non-spontaneously.

Statement II: A process in which all steps cannot be retraced by themselves is called a spontaneous process.

- (1) If both the statements are true and statement-II is the correct explanation of Statement-I
- (2) If both the statements are true but statement-II is not the correct explanation of Statement-I
- (3) If Statement-I is true and Statement-II is false.
- (4) If Statement-I is False and Statement-II is true

BIOLOGY

91. The process of photo-phosphorylation take place in
 (1) Chloroplast (2) Ribosomes
 (3) Mitochondria (4) Cell-wall

92. Match the following columns :

mn-I	mn-II
Plant growth oters	n
Plant growth itor	erellin
	isic acid
	kinin

- (1) a-i, ii, iii, b-iv
 (2) a-iv, b-i, ii, iii
 (3) a-iii, b-i, ii, iv
 (4) a-i, ii, iv, b-iii

93. Match the growth parameters (Column I) with their correct descriptions (Column II).

Column I (Growth Parameter)	Column II (Description)
h measurement	red using increase in area (e.g., leaf expansion)
er of cells	tes increase in cell count through division
or dry weight	red using weight increment due to growth
ee area	red using increase in length (e.g., root or stem)

- (1) A-(iv), B-(ii), C-(iii), D-(i)
 (2) A-(i), B-(ii), C-(iv), D-(iii)
 (3) A-(iii), B-(iv), C-(ii), D-(i)
 (4) A-(ii), B-(iii), C-(iv), D-(i)

94. Which one of the following is true about photorespiration?
 (1) It is found in mesophyll cells of C_4 plants.
 (2) Neither CO_2 nor NADH are formed
 (3) It takes place in high CO_2 and low O_2
 (4) Neither sugar nor ATP are synthesized

95. Which of the following are true –
 (a) F_0 of ATP synthase is a integral membrane protein complex
 (b) Passage of protons through F_0 channel is coupled to the catalytic site of the F_1 component for the production of ATP
 (c) Oxidation of NADPH to NAD^+ is very vigorous in fermentation
 (d) F_1 headpiece is a integral membrane protein complex
 (1) Both (a) and (c)
 (2) (a), (b), (c) and (d)
 (3) (a), (b) and (d)
 (4) Both (a) and (b)

96. Which of the following are produced by reactions that take place in the thylakoids and consumed by reactions in the biosynthetic phase:

- (1) ATP and NADH
 (2) $NADP^+$ and ADP
 (3) Glucose and O_2
 (4) ATP and NADPH

97. Which of the following is **incorrect** regarding the Krebs's cycle:

- (1) It is also known as Tricarboxylic Acid Cycle.
 (2) It occurs in mitochondria.
 (3) The TCA cycle starts with the condensation of acetyl group with oxaloacetic acid (OAA) and water to yield citric acid.
 (4) It does not involve any decarboxylation step.

98. Cytochrome C oxidase complex contains:
 (1) Cytochrome a, a_3 and one copper centre
 (2) Cytochrome a and two copper centres
 (3) Cytochrome a_3 and one copper centre
 (4) Cytochrome a, a_3 and two copper centre

99. Which of the following is not a product of light reaction of photosynthesis?

- (1) NADPH
 (2) NADH
 (3) ATP
 (4) Oxygen

100. PGA as the first carbon dioxide fixation product which was discovered in photosynthesis of algae. It the first stable compound in which type of plants

- (1) Only algae
- (2) Some algae and bryophytes
- (3) In all green plants
- (4) In all C_3 green plants

101. What causes a green plant to bend towards light as it grows?

- (1) Because green plants need light to carry on photosynthesis
- (2) Because green plants are phototropic
- (3) Light stimulates plant cells on the lighted side to grow faster
- (4) Auxin accumulates on the shaded side stimulating greater cell elongation.

102. Which is the incorrect about auxin?

- (1) Proceeds flowering in pineapple
- (2) Extracted from human urine
- (3) It induces parthenocarp in tomato
- (4) 2, 4-D is natural auxin.

103. According to chemiosmotic theory of P. Mitchell (1978), ATP are synthesized on membranes due to the:

- (1) Proton gradient
- (2) Electron gradient
- (3) Osmosis
- (4) From H_2SO_4

104. Splitting of water molecules during the light reaction of photosynthesis is associated with:

- A. PS I resulting in the release of O_2 , protons and transfer of electrons to PS II
- B. PS II resulting in the release of O_2 , protons and transfer electrons to PS I
- C. PS II resulting in the release of O_2 , protons and transfer to electrons to PS II
- D. PS I resulting in the release of O_2 , protons and transfer of electrons to PSI

- (1) A is correct (2) B is correct
- (3) C is correct (4) D is correct

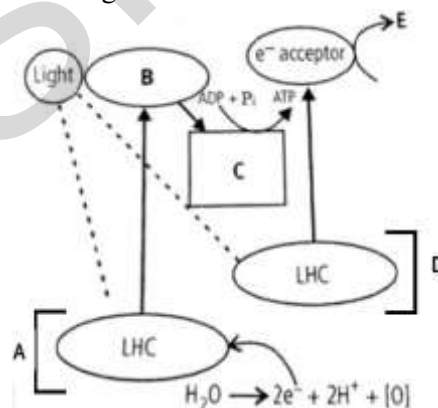
105. Pyruvate dehydrogenase complex is used in converting -

- (1) Pyruvate to glucose
- (2) Glucose to pyruvate
- (3) Pyruvic acid to lactic acid
- (4) Pyruvate to acetyl CO-A

106. In the eukaryotes, ...A... takes place within the chloroplasts, whereas the breakdown of complex molecules to yield energy takes place in the ...B... and in the ...C...

- (1) A- photosynthesis, B- cytoplasm, C-mitochondria
- (2) A-respiration, B-cytoplasm, C-mitochondria
- (3) A- respiration, B- chloroplast, C- cytoplasm
- (4) A-photosynthesis, B-chloroplast, C-cytoplasm

107. Identify A, B, C, D and E in the flowchart showing Z-scheme of light reaction.



	A	B	C	D	E
(1)	P_{700}	H^+ acceptor	e^- acceptor	P_{680}	$NADP^+$
(2)	Photosystem I	e^- acceptor	e^- transport	Photosystem II	NADPH
(3)	Photosystem II	H^+ Acceptor	e^- transport	P_{700}	NADPH
(4)	Photosystem II	e^- acceptor	e^- acceptor	Photosystem I	NADPH

108. Why Calvin cycle is called C_3 -cycle?

- (1) Primary CO_2 acceptor is C_3 compound
- (2) Many intermediate compounds are C_3 compound.
- (3) 1st stable product is 3 PGA which is a three carbon compound.
- (4) 1st stable product is 3-PGAL which is C_3 compound.

- 109.** The absorption spectrum of chlorophyll-
- (1) Shows that some colours of light are absorbed more than the others
 - (2) Approximates action spectrum of photosynthesis
 - (3) Explains why chlorophyll is a green pigment
 - (4) Has all the above properties

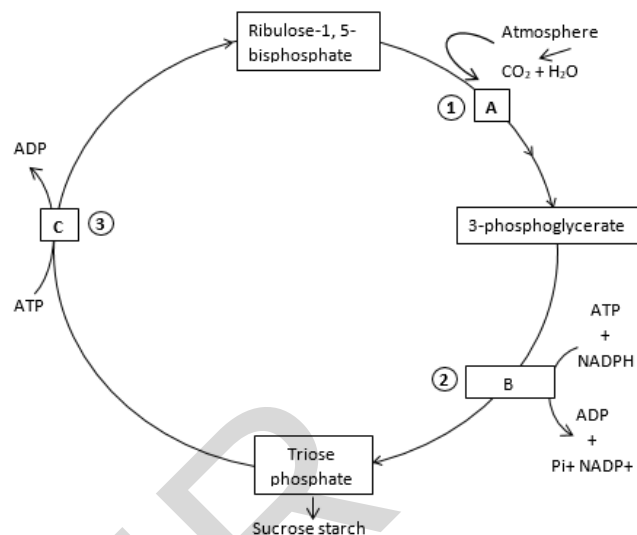
- 110.** Read the following statements carefully and choose the correct option:
- (i) Auxins promote cell elongation and maintain apical dominance in shoots.
 - (ii) Gibberellins cause bolting and breaking of seed dormancy.
 - (iii) Cytokinins delay senescence and promote cell division.
 - (iv) Ethylene promotes fruit ripening and epinasty but inhibits elongation growth.
 - (v) Absciscic acid (ABA) acts as a growth inhibitor and induces seed dormancy and stomatal closure.
- (1) Only i, ii and iii are correct
 - (2) Only i, ii and iv are correct
 - (3) Only i, ii, iii and v are correct
 - (4) All i, ii, iii, iv and v are correct

- 111.** Match the List I with List II & select correct answer

	List I (Mitochondrial component)		List II (Complex)
A.	ATP synthase	i.	Complex-I
B.	FMN-NADH ₂ dehydrogenase	ii.	Complex-II
C.	CoQ-FADH ₂ dehydrogenase	iii.	Complex-III
D.	Cyto.a-cyto.a ₃	iv	Complex-IV
E.	Cyto.b-cyto.c1	v.	Complex-V

- (1) A-v, B-i, C-iv, D-ii, E-iii
- (2) A-v, B-i, C-iii, D-ii, E-iv
- (3) A-v, B-i, C-ii, D-iv, E-iii
- (4) A-v, B-ii, C-i, D-iv, E-iii

- 112.** Identify A, B and C in the given figure and choose the correct option from the set (A-C) given below.



- (1) A-Reduction, B-Carboxylation, C-Regeneration
- (2) A-Reduction, B-Regeneration, C-Carboxylation
- (3) A-Carboxylation, B-Reduction, C-Regeneration
- (4) A-Carboxylation, B-Regeneration, C-Reduction

- 113.** I. Tropical plants have a.....A.....than plants adapted to temperate climates.
 II. Water has a/an.....B.....effect on the rate of photosynthesis.
 III. The optimum temperature for photosynthesis of different plants depends onC.....
 Identify A, B and C choose the **correct** option.

	A	B	C
(1)	Higher temperature optimum	Direct	Habitat
(2)	Lower temperature optimum	Indirect	Water
(3)	Lower temperature optimum	Direct	Light
(4)	Higher temperature optimum	Indirect	Habitat

- 114.** Most of the energy of the carbohydrates is released by oxidation when
- (1) Pyruvic acid is converted into CO₂ and H₂O
 - (2) Pyruvic acid is converted into acetyl Co-A
 - (3) Sugar is converted into pyruvic acid
 - (4) Glucose is converted into alcohol and CO₂

115. If RQ is less than 1 in a respiratory metabolism, it would mean that

- (1) Carbohydrates are used as respiratory substrate
- (2) Organic acids are respiratory substrate
- (3) The oxidation of the respiratory substrate consumed more oxygen than the amount of CO_2 released
- (4) The oxidation of the respiratory substrate consumed less oxygen than the amount of CO_2 released

116. Consider the following statements.

- I. In biosynthetic phase (C_3 cycle), enzymes are present in the matrix of Golgi body.
- II. C_3 and C_4 cycle are two parts of biosynthetic phase of photosynthesis in C_3 plants.

- (1) Both I and II are true
- (2) Both I and II are false
- (3) I is true, II is false
- (4) I is false. II is true

117. Which one of these statements is incorrect?

- (1) Glycolysis operates as long as it is supplied with NAD^+ that can pick up hydrogen atoms
- (2) Glycolysis occurs in cytosol
- (3) Enzymes of TCA cycle are present in mitochondrial matrix
- (4) Oxidative phosphorylation takes place in outer mitochondrial membrane

118. Select the correct combination:

- (1) GA_3 - responsible for immature falling of leaves
- (2) Cytokinin - overcome necrosis
- (3) GA-Induce flowering in Pineapple
- (4) 2, 4 D- stem elongation in sugar cane

119. An auxin which is widely used to kill the dicotyledonous weed is

- (1) IAA
- (2) IBA
- (3) NAA
- (4) 2, 4-D

120. Match the column I and II and select the correct code:-

	Column-I		Column-II
A.	Control on weeds	i.	GA_3
B.	Induction of germination	ii.	CK
C.	Ripening of Fruit	iii.	2,4-D
D.	Delaying of senescence	iv.	C_2H_4

- (1) A-ii, B-iv, C-iii, D-i
- (2) A-iii, B-i, C-iv, D-ii
- (3) A-i, B-ii, C-iv, D-iii
- (4) A-ii, B-iv, C-i, D-iv

121. Match the following Column-I and Column-II.

Column-I (Hormones)	Column-II (Function)
(A) IAA	(i) Herring sperm DNA
(B) ABA	(ii) Increasing length of grape's stalk
(C) Ethylene	(iii) Stomatal closure
(D) GA	(iv) Weed- free lawns
(E) Cytokinin's	(v) Ripening of fruits

- | | | | | |
|--------|-----|----|-----|----|
| A | B | C | D | E |
| (1) iv | iii | v | ii | i |
| (2) v | iii | iv | ii | i |
| (3) iv | i | v | iii | ii |
| (4) v | iii | ii | i | iv |

122. Ribulose Bisphosphate carboxylase enzyme, catalyze the carboxylation reaction between

- (1) CO_2 and ribulose-1, 5-Bisphosphate
- (2) Oxaloacetic acid and acetyl Co-A
- (3) PGA and dihydroxyacetone phosphate
- (4) Ribulose Bisphosphate and phosphate glyceraldehyde

123. Select the wrong pair

(1)	GA_3	Acidic
(2)	IAA	Indole compound
(3)	Cytokinins	Carotenoid derivatives
(4)	ABA	General plant growth inhibitor

124. Plant hormone associated with germination of peanut seeds is -

- (1) Auxin
- (2) Cytokinin
- (3) Ethylene
- (4) Gibberellin

125. Glycolysis give rise to

- (1) 8ATP, 2NADH.H^+ Pyruvate
- (2) 2ATP, 2CoA , 2NADH.H^+
- (3) 2ATP, 2NADH.H^+ , 2 Pyruvate
- (4) 2ATP, 2 Acetate, 2NADPH.H^+

126. Match column I with column II and select the **correct** option from the codes given below:

	Column-I		Column-II
A.	Glycolysis	i.	Glycogen
B.	Complex-V	ii.	ATP Synthase
C.	Storage form of glucose	iii.	EMP pathway
D.	Common respiratory substrate of glycolysis	iv	Glucose

- (1) A-(iv), B-(ii), C-(i), D-(iii)
 (2) A-(iii), B-(ii), C-(i), D-(iv)
 (3) A-(iv), B-(iii), C-(i), D-(ii)
 (4) A-(i), B-(ii), C-(iii), D-(iv)
127. Select the **incorrect** option.
- (1) Primary growth is the increase in girth of axis due to activity of apical meristem
 (2) Secondary growth is the increase in girth due to activity of lateral meristem (vascular cambium and cork cambium)
 (3) Elongation of plant is the primary growth
 (4) Both (2) and (3)
128. Van Niel demonstrated that photosynthesis is essentially a light dependent reaction in which hydrogen from a suitable oxidisable compound reduces carbon dioxide to carbohydrates. This study is based on the organisms of :
- (1) Only purple bacteria
 (2) Only green sulphur bacteria
 (3) Chlorella and Chlamydomonas
 (4) Purple and green sulphur bacteria
129. Growth in plant is measured by the increase in:
- A. Dry weight
 B. Length, area and volume
 C. Number of Cell
 D. Cell division in per unit time
 E. Fresh weight
 F. Size of cell
- Choose the correct option:
- (1) 5 (2) 4
 (3) 2 (4) 1

130. Which of the following steps during glycolysis is associated with production of ATP?
- (1) Glucose $\rightarrow$ Glucose – 6 – phosphate.
 (2) Fructose-6-phosphate $\rightarrow$ Fructose-1, 6-biphosphate.
 (3) PEP $\rightarrow$ Pyruvic acid.
 (4) Both (1) and (2).
131. Which of the following is correct option :
- (1) Auxin help to initiate rooting in stem cutting.
 (2) Ethylene reduces root growth.
 (3) Ethylene promoted root hair formation.
 (4) Both (1) and (3).
132. Substrate level phosphorylation occurs in glycolysis when there is conversion of
- (1) BPGA $\rightarrow$ PGA and phosphoglycerate $\rightarrow$ PEP
 (2) BPGA $\rightarrow$ PGA and PEP $\rightarrow$ Pyruvic acid
 (3) Fructose-6-phosphate $\rightarrow$ Fructose 1, 6-biphosphate
 (4) Glucose $\rightarrow$ Glucose-6-phosphate
133. **Assertion (A):** 6 cycles of Calvin cycle are required for synthesis of one glucose molecule.
Reason (R): In each Calvin cycle, only one molecule of CO_2 is fixed. Hence, 6 molecules of CO_2 are required for one molecule of glucose.
- (1) If both A and R are true and R is the correct explanation of A
 (2) If both A and R are true but R is not the correct explanation of A
 (3) If A is true, but R is false
 (4) If A is false, but R is true
134. Which of the following is true regarding glycolysis?
- I. Generate 4 ATP directly by substrate level phosphorylation
 II. Produces no ATP
 III. Has no connection with electron transport chain.
 IV. Reduces two molecules of NAD^+ for every glucose molecule processed.
- (1) Only I
 (2) I, II and III
 (3) I and II
 (4) None of these

- 135.** Conditions necessary for photosynthesis are:
 (1) Light and suitable temperature
 (2) Chlorophyll and water
 (3) Carbon dioxide
 (4) All of the above
- 136.** During photochemical reaction of photosynthesis -
 (1) liberation of O_2 takes place
 (2) Formation of ATP and NADPH take place
 (3) Liberation of O_2 , formation of ATP, and NADPH takes place
 (4) Assimilation of CO_2 takes place
- 137.** The electron ejected by P_{680} in light reaction is initially accepted by
 (1) Phaeophytin (2) ATP
 (3) Ferredoxin (4) P-700
- 138.** How many ATP and $NADPH.H^+$ are required for the fixation of one CO_2 in C_3 cycle?
 (1) 2 ATP & 2 NADPH
 (2) 3 ATP & 3 NADPH
 (3) 3 ATP & 2 NADPH
 (4) 2 ATP & 3 NADPH
- 139.** Photosynthesis is often described as an anabolic, endergonic process driven by light energy. Consider the following statements related to this process:
 (i) The overall process of photosynthesis represents a reduction of CO_2 into carbohydrates and oxidation of water into O_2 .
 (ii) The equation $6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$ correctly represents oxygenic photosynthesis.
 (iii) Photosynthesis is the exact reverse of respiration in both reactants and mechanism.
 (iv) The light reaction provides assimilatory power (ATP and $NADPH_2$) required for CO_2 fixation in the dark reaction.
 Select the most appropriate option:
 (1) Statements (i), (ii), and (iv) are correct.
 (2) Only statements (i) and (iii) are correct.
 (3) All statements are correct.
 (4) Statements (ii) and (iii) are correct.
- 140.** The first stable product of photosynthesis in C_4 plants is:
 (1) Oxaloacetic acid
 (2) 3-PGA
 (3) Fructose-1, 6-diphosphate
 (4) Ribulose-1, 5-di phosphate
- 141.** T.W. Engelmann using a prism and split the light into its spectral components and then illuminated a green alga, placed in a suspension of aerobic bacterium. Name that green alga :
 (1) Chlorella
 (2) Cladophora
 (3) Chlamydomonas
 (4) Nostoc and Anabaena
- 142.** The plant hormone, which is synthesized in root and is translocated to aerial parts is:
 (1) Auxin
 (2) Cytokinin
 (3) Traumatins
 (4) Morphactins
- 143.** Nutrient mobilization for delaying senescence in leaves is due to
 (1) Cytokinin (2) Gibberellin
 (3) Auxin (4) Ethylene
- 144.** Which of the following begins the Calvin cycle and is the commitment step that results the entire pathway being carried out?
 (1) $3PGA \xrightarrow{ATP, NADPH_2} 3PGAL$
 (2) The regeneration of RuBP
 (3) $CO_2 + RuBP \rightarrow 3PGA$
 (4) It can start from anywhere
- 145.** Select the **incorrect** statement :-
 (1) The increased growth per unit time is termed as growth rate.
 (2) Growth is expressed in increase in cell number.
 (3) Arithmetic growth rate plot a linear curve and it expressed as $L_t = L_0 - rt$.
 (4) Geometrical growth rate plot a curve is sigmoid and expressed as $W_t = W_0 e^{rt}$.

146. Kinetin was discovered by:
 (1) Miller (2) Letham
 (3) Benson & Calvin (4) Thiemann & Went
147. In Calvin cycle what is the **correct** sequence of substances:
 (1) A ketose sugar, 3-phosphoglycerate, Triose phosphate
 (2) 3-C organic acid, A ketose sugar, Triose phosphate
 (3) Triose phosphate, 3-C organic acid, A ketose sugar
 (4) Ribose 1, 3 biphosphate, 3- phosphoglycerate, triose phosphate
148. The sequential stages of Calvin cycle are:
 (1) Carboxylation, Reduction and Regeneration
 (2) Carboxylation, Regeneration and Reduction
 (3) Reduction, Carboxylation and Regeneration
 (4) Regeneration, Reduction and Carboxylation.
149. Identify the correct statement in relation to C₄ plants:
 (1) Kranz anatomy is an essential feature for C₃ plants.
 (2) C₃ plants have higher water use efficiency than C₄ plants.
 (3) Conversion of oxaloacetate to malate occurs in the bundle sheath cell.
 (4) Photo respiration can be minimized where C₄ pathway is in operation.
150. During non-cyclic photophosphorylation, electrons are continuously lost from the reaction centre of PS-II. Which source is used to replace these electrons:
 (1) NADP (2) ATP
 (3) PS-II (4) Water
151. Which statement is false about photosynthesis:
 (1) The electron carriers involved in photophosphorylation are located in the thylakoid membranes.
 (2) In the photosynthetic process, PS-II absorbs energy at just below 680 nm.
 (3) The absorption spectrum of chlorophyll shows that some colours of light are absorbed more than the others.
 (4) Compensation point of C₄ plant is higher than C₃ plants.
152. Final decarboxylation in Krebs cycle occur between—
 (1) Citric acid and α -ketoglutaric acid
 (2) α - ketoglutaric acid and succinic acid
 (3) Succinic acid and malic acid
 (4) Malic acid and oxalo acetic acid
153. During conversion of succinyl Co-A to succinic acid a molecule of GTP is formed, this process is known as: -
 (1) Oxidative phosphorylation
 (2) Substrate level phosphorylation
 (3) Photophosphorylation
 (4) Terminal oxidation
154. Which of the following elements is an/activator for both ribulose biphosphate carboxylase oxygenase and phosphoenol pyruvate carboxylase in photosynthetic carbon fixation?
 (1) Mg^{2+} (2) Zn^{2+} (3) Ca^{2+} (4) SO_4^{2-}
155. Select the **wrong** statement
 (1) When tripalmitin is used as a substrate in respiration, the RQ is 0.7.
 (2) One glucose molecule yields a net gain of 2 ATP molecules during fermentation.
 (3) The intermediate compound which links glycolysis with Kreb's cycle is malic acid.
 (4) One glucose molecule yields a net gain of 38 ATP molecules during aerobic respiration.
156. The cells positioned away from the root apical meristems differentiate as....A...., while those pushed to the periphery mature asB.....
 (1) A-root cap cells, B-epidermis
 (2) A-endodermis, B-epidermis
 (3) A-root cap cells, B-endodermis
 (4) A-epidermis, B-endodermis
157. Which of the metabolites is common to respiration mediated breakdown of fats, carbohydrates and proteins?
 (1) Glucose -6-phosphate
 (2) Fructose 1-6- biphosphate
 (3) Pyruvic acid
 (4) Acetyl co-A
158. In mitochondrial ETS, cytochrome 'C' acts as a mobile carrier for transfer of electrons between:
 (1) Complex III and IV
 (2) Complex I and II
 (3) Complex II and III
 (4) Complex IV and V

159. Match the plant hormones given in column-I with their function/other name given in column-II and choose the correct combination.

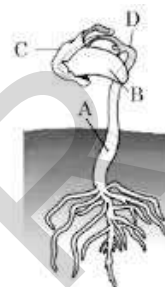
	Column-I (plant hormone)		Column-II (Function/other name)
a.	Zeatin	I.	Flowering hormone
b.	Florigen	II.	Synthetic auxin
c.	IBA	III.	Cytokinin
d.	NAA	IV.	Natural auxin

- (1) a - III, b - IV, c - I, d - II
 (2) a - II, b - I, c - IV, d - III
 (3) a - I, b - II, c - III, d - IV
 (4) a - III, b - I, c - IV, d - II
160. Select the correct pathway for electron transport during photosynthesis.
 (1) $\text{CO}_2 \rightarrow \text{RuBP} \rightarrow \text{Glucose} - \text{ATP}$
 (2) $\text{H}_2\text{O} \rightarrow \text{PS-I} \rightarrow \text{PS-II} \rightarrow \text{NADPH} + \text{H}^+$
 (3) $\text{H}_2\text{O} \rightarrow \text{PS-II} \rightarrow \text{PS-I} \rightarrow \text{NADP}^+$
 (4) $\text{H}_2\text{O} \rightarrow \text{PS-II} \rightarrow \text{PS-I} \rightarrow \text{ATP}$
161. Choose the correct combination of A and B.
 The NADH.H^+ synthesised in...A...is transferred into the mitochondria and undergoes oxidative ...B...
 (1) A- EMP; B-Carboxylation
 (2) A- ETS; B- Phosphorylation
 (3) A- Glycolysis; B-Phosphorylation
 (4) A- TCA cycle; B- Decarboxylation
162. Find the correct statement(s) from the following statements :
 (a) In plants light saturation occurs at 10% of the moderate sunlight.
 (b) The concentration of CO_2 , in atmosphere is 0.03-0.04 percent.
 (c) C_3 plants show saturation to CO_2 concentration at 360 mL.L^{-1} .
 (d) C_4 plants show saturation to CO_2 , beyond 450 mL.L^{-1} .
 (1) Only (a), (d) (2) Only (b)
 (3) Only (d) (c) (4) All
163. **Statement-I:** Auxin promotes formation of abscission layer in old leaves & fruits.
Statement-II: Auxin prevents abscission in younger leaves & fruits.
 (1) Statement I and Statement II are correct
 (2) Statement I and Statement II are incorrect
 (3) Statement I is correct but Statement II incorrect
 (4) Statement I is incorrect but Statement II is correct

164. Which of the following is not related with Gibberellins?

- (1) Speed up malting process in brewing industry
 (2) Spraying sugarcane crop with gibberellins increases the length of the stem.
 (3) Spraying juvenile conifers with GAs hastens the maturity period.
 (4) Helps in lateral shoot growth.

165. Identify A, B, C and D in the given figure and choose the correct option.



- (1) A-Hypocotyl, B-Cotyledon, C-Seed coat, D-Epicotyl hook
 (2) A-Epicotyl, B-Cotyledon, C-Hypocotyl, D-Seed coat
 (3) A- Epicotyl, B- Seed coat, C- Hypocotyl, D- Cotyledon
 (4) A-Hypocotyl, B-Seed coat, C-Epicotyl, D- Cotyledon
166. In which of the following light is limiting factor
 (1) Surface dwellers
 (2) During cloudy day
 (3) During night
 (4) Both (1) and (2)
167. Photorespiration is a waste process because
 (1) it consume ATP
 (2) No sugar synthesis
 (3) loss of carbon
 (4) All of these
168. In addition to the 12 molecules of NADPH.H^+ the energy required for the synthesis of one mole of hexose by C_3 and C_4 pathway is
 (1) 18 molecules of ATP
 (2) 30 molecules of ATP
 (3) 18 & 30 molecules of ATP respectively
 (4) 30 & 18 molecules of ATP respectively

- 169.** Which of the following process does not release CO_2 ?
 (1) Alcoholic fermentation
 (2) Oxidative decarboxylation of pyruvic acid.
 (3) Conversion of oxalo-succinic acid to α -Keto-glutaric acid
 (4) Lactic acid fermentation
- 170.** How many ATP generate through substrate level phosphorylation in a Krebs cycle?
 (1) 4 (2) 1 (3) 3 (4) 5
- 171.** Which feature is not found in C_4 plants?
 (1) Presence of RuBisCO in Mesophyll cells.
 (2) Kranz anatomy
 (3) Carboxylation proceeds two times
 (4) The optimum temperature for photosynthesis is 40°C .
- 172.** **Statement I :** Cellular respiration is an amphibolic process.
Statement II: The carbon skeleton (intermediates of respiration) produced during respiration is used as precursor for biosynthesis of other molecules in cell.
 (1) Both Statements I & II are incorrect
 (2) Statement I is correct but Statement II incorrect
 (3) Both Statement I and Statement II are correct
 (4) Statement I is incorrect but Statement II is correct
- 173.** Which of the following is the energy yielding step of glycolysis?
 (1) Glucose $\rightarrow$ Glucose-6-phosphate
 (2) BPGA $\rightarrow$ PGA
 (3) Fructose 1, 6 biphosphate $\rightarrow$ PGAL
 (4) Phosphoglycerate $\rightarrow$ phosphoenolpyruvate
- 174.** Which of the following is primary CO_2 acceptor in CAM plants.
 (1) RUBP (2) OAA
 (3) Malic acid (4) PEP
- 175.** Which of the following is final electron acceptor in ETS
 (1) Hydrogen (2) Oxygen
 (3) Nitrogen (4) Water
- 176.** Statement-I Aerobic respiration is a redox reaction
 Statement-II Anaerobic respiration is a oxidation reaction
 (1) Both Statements I & II are correct
 (2) Statement I is correct but Statement II incorrect
 (3) Both Statement I and Statement II are incorrect
 (4) Statement I is incorrect but Statement II is correct
- 177.** Assertion : Photosynthesis is a uphill process
 Reason : CO_2 is a weak electron acceptor
 (1) If both A and R are true and R is the correct explanation of A
 (2) If both A and R are true but R is not the correct explanation of A
 (3) If A is true, but R is false
 (4) If A is false, but R is true
- 178.** During ETS the movement of electrons is always
 (1) Uphill (2) Downhill
 (3) Both (1) and (2) (4) None of these
- 179.** Which of the following increase the rate of metabolism during malting
 (1) Auxin (2) Cytokinin
 (3) Gibbrellin (4) ABA
- 180.** Respiration is a
 (1) Anabolic process
 (2) Catabolic process
 (3) Amphibolic process
 (4) None of these

Answer Key and Solution

ANSWER KEY & SOLUTION

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	3	4	1	3	1	1	4	4	3	3	4	4	3	1
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	2	1	1	2	1	1	1	3	1	4	3	3	1	2	1
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	1	2	3	1	3	1	2	1	2	3	2	3	4	2	2
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	4	2	2	3	4	1	4	3	1	3	3	3	1	2	1
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	3	1	3	4	1	3	3	1	3	4	4	4	1	4	1
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	3	2	3	2	3	4	1	3	4	3	1	1	3	3	4
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	1	4	1	4	4	4	4	4	2	4	4	4	1	2	4
Que.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Ans.	1	4	3	4	4	3	3	4	1	3	2	4	2	4	2
Que.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Ans.	1	1	3	3	3	2	1	4	1	3	4	2	1	1	4
Que.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Ans.	3	1	3	1	1	2	2	1	3	3	1	1	1	4	4
Que.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
Ans.	4	2	2	1	3	1	4	1	4	3	3	2	1	4	1
Que.	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
Ans.	4	2	3	4	2	1	3	2	4	2	1	1	2	3	3

PHYSICS

1. (2)

Sol. Stress = $Y \times \text{strain}$

$$\frac{Mg}{A} = Y(\text{strain})$$

$$M = \frac{2 \times 10^{11} \times 10^{-3} \times 3 \times 10^{-6}}{10}$$

$$M = 60 \text{ kg}$$

2. (3)

$$\text{Sol. } \Delta \ell = \frac{F \ell}{(\pi r^2) Y} \quad \ell$$

$$\Delta \ell \Rightarrow \frac{\Delta \ell}{\ell} = \frac{W}{4W} = 1$$

$$\Delta \ell = \ell$$

3. (4)

$$\text{Sol. As, } PV = \frac{m}{M} RT$$

$$\therefore \text{Initially, } PV = \frac{6}{M} R \times 500 \quad \dots(i)$$

$$\text{and finally, } \frac{P}{2} V = \frac{(6-x)}{M} R \times 300$$

(if x g of gas leaks out) ... (ii)

$$\text{Dividing eqn. (i) by (ii), we get } 2 = \frac{6}{6-x} \times \frac{5}{3}$$

On solving, we get $x = 1 \text{ g}$

4. (1)

Sol. Number of moles of gas in two flasks are

$$n_1 = \frac{P_1 V_1}{RT} \quad \text{and} \quad n_2 = \frac{P_2 V_2}{RT} \quad \therefore n = n_1 + n_2$$

$$\therefore P = \frac{(n_1 + n_2) RT}{V_1 + V_2} = \frac{P_1 V_1 + P_2 V_2}{V_1 + V_2}$$

5. (3)

$$\text{Sol. } W = 4\pi R^2 T [n^{1/3} - 1]$$

$$= 4\pi R^2 T [(8)^{1/3} - 1] = 4\pi R^2 T$$

6. (1)

$$\text{Sol. } P_1 = 3P_2$$

$$\frac{4T}{r_1} = 3 \times \frac{4T}{r_2}; \frac{r_1}{r_2} = \frac{1}{3} \Rightarrow \frac{A_1}{A_2} = \left(\frac{r_1}{r_2}\right)^2 = 1:9$$

7. (1)

Sol. Water rise up to the top of capillary tube and stays there without overflowing.

8. (4)

Sol. Here, $P_1 = 4 \text{ atm}$, $T_1 = 27^\circ\text{C} = 300 \text{ K}$,

$$V_1 = 1500 \text{ m}^3, P_2 = 2 \text{ atm}, T_2 = -3^\circ\text{C} = 270 \text{ K},$$

$$V_2 = ?$$

$$\text{As } \frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\therefore V_2 = \frac{P_1 V_1 T_2}{T_1 P_2} = \frac{4 \times 1500 \times 270}{300 \times 2} = 2700 \text{ m}^3$$

9. (4)

$$\text{Sol. } h \propto \frac{T}{\rho} \Rightarrow \frac{h_1}{h_2} = \frac{T_1}{T_2} \times \frac{\rho_2}{\rho_1}; \frac{h_1}{h_2} = \frac{60}{50} \times \frac{0.6}{0.8} = \frac{9}{10}$$

10. (3)

$$\text{Sol. } V_T \propto r^2$$

$$\frac{V_1}{V_2} = \left(\frac{R}{3R}\right)^2 = 1:9$$

11. (3)

Sol. Water has maximum density at 4°C , so if the water is heated above 4°C cooled below 4°C density decreased, i.e., volume increases.

12. (4)

Sol. Amount of heat lost by water = Amount of heat gained by ice.

$$\Rightarrow m_w S_w (\Delta T)_w = m_i L_f + m_i S_w (\Delta T)_i$$

$$\Rightarrow 100 \times 1 \times (50 - T) = 10 \times 80 + 10 \times 1 \times (T - 0)$$

$$\Rightarrow 5000 - 100 T = 800 + 10 T$$

$$\Rightarrow 11 T = 420 \text{ or } T = 38.2^\circ\text{C}, 4200 = 110 T$$

13. (4)

Sol. Due to large specific heat of water, it release large heat with very small temperature change.

14. (3)

Sol. With rise of altitude pressure decrease and boiling point decrease.

15. (1)

Sol. $d_A = 2 \text{ cm}$ and $d_B = 4 \text{ cm}$

$\therefore r_A = 1 \text{ cm}$ and $r_B = 2 \text{ cm}$

From equation of continuity, $av = \text{contact}$

$$\therefore p \left(\frac{d_A}{2} \right)^2 v_A = p \left(\frac{d_B}{2} \right)^2 v_B$$

$$\frac{d_A^2}{4} v_A = \frac{d_B^2}{4} v_B$$

$$d_A^2 v_A = d_B^2 v_B$$

$$\frac{v_A}{v_B} = \left(\frac{d_B}{d_A} \right)^2$$

$$\frac{v_A}{v_B} = 4$$

$$\Rightarrow v_A = 4v_B$$

16. (2)

Sol. $V_{\text{rms}} \propto \frac{1}{\sqrt{M}}$

17. (1)

Sol. $(C_V)_{\text{mix}} = \frac{\mu_1 C_{V_1} + \mu_2 C_{V_2}}{\mu_1 + \mu_2}$

$$(C_V)_{\text{mix}} = \frac{1}{2} \left[\frac{3}{2} R + \frac{5}{2} R \right] = 2R$$

$$C_P - C_V = R \Rightarrow C_P = C_V + R = 2R + R = 3R$$

$$\gamma = \frac{C_P}{C_V} = \frac{3R}{2R} = \frac{3}{2} \Rightarrow \gamma = \frac{3}{2}$$

18. (1)

19. (2)

20. (1)

Sol. According to ideal gas equation

$$PV = nRT \text{ or } \frac{V}{T} = \frac{nR}{P}$$

At constant pressure

$$\frac{V}{T} = \text{constant}$$

21. (1)

Sol. Free expansion (No external process)

So, $W = 0$

No heat exchange $Q = 0$

$$\Delta U = 0$$

22. (1)

Sol. Mean free path, $\lambda = \frac{1}{\sqrt{2} \pi d^2 n}$

where,

n = Number of molecules per unit volume

d = Diameter of the molecules

$$\text{As } PV = k_B n T \therefore n = \frac{N}{V} = \frac{P}{k_B T} \therefore \lambda = \frac{k_B T}{\sqrt{2} \pi d^2 P}$$

Where, k_B = Boltzmann constant

P = Pressure of gas

T = Temperature of the gas

23. (3)

24. (1)

Sol. $v_A = v_B$ (since area is uniform)

From Bernoulli's principle

$$\frac{v_A^2}{2} + gh + \frac{p_A}{\rho} = \frac{v_B^2}{2} + 0 + \frac{p_B}{\rho}$$

$$\Rightarrow p_B > p_A$$

25. (4)

Sol. $P \propto T^3$

For an adiabatic process

$$PT^{\frac{\gamma}{\gamma-1}} = \text{constant} = \frac{\gamma}{\gamma-1} \rightarrow \gamma = 1.5$$

26. (3)

Sol. Here, $V_1 = V$, $V_2 = 2V_1 = 2V$, $T_1 = 27^\circ\text{C}$
 $= 300 \text{ K}$

According to Charles' law $\frac{V_1}{T_1} = \frac{V_2}{T_2}$

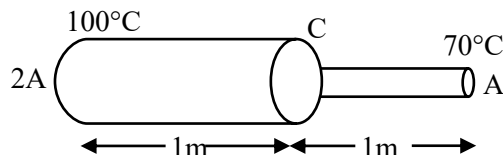
$$\text{or } T_2 = T_1 \left(\frac{V_2}{V_1} \right) = 300 \text{ K} \left(\frac{2V}{V} \right)$$

$$= 600 \text{ K} = 327^\circ\text{C}$$

Increase in temperature $\Delta T = T_2 - T_1$

$$= (327 - 27)^\circ\text{C} = 300^\circ\text{C}$$

27. (3)



Sol.

$$K(2A)(100 - T) = KA(T - 70)$$

$$200 - T = T - 70$$

$$3T = 270$$

$$T = 90^\circ\text{C}$$

28. (1)

Sol. When the temperature of the atmosphere reaches below 0°C , then the water vapours present in air, instead of condensing, freeze directly in the form of minute particles of ice. Many particles coalesce and take cotton-like shape which is called snow. Thus snow contains air packets in which convection currents cannot be formed. Hence snow is a good heat insulator. In ice there is no air.

29. (2)

Sol. Heat energy required to just melt the bullet
 $Q = Q_1 + Q_2$
 Here, $Q_1 = ms\Delta T = m \times 125 \times (327 - 27) = 3.75 \times 10^4 \text{ m}$
 $Q_2 = mL = m \times 2.5 \times 10^4 = 2.5 \times 10^4 \text{ m}$
 $\therefore Q = 3.75 \times 10^4 \text{ m} + 2.5 \times 10^4 \text{ m}$
 $Q = 6.25 \times 10^4 \text{ m}$

Let v be the speed of bullet, then 50% of $\frac{1}{2}mv^2$ should be equal to Q .

$$\therefore 50\% \times \frac{1}{2}mv^2 = Q$$

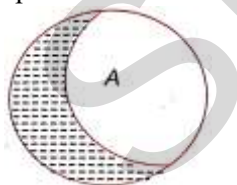
$$\therefore 0.5 \times \frac{1}{2}mv^2 = 6.25 \times 10^4 \text{ m}$$

$$\Rightarrow v = \sqrt{\frac{6.25 \times 10^4 \times 2}{0.5}} = 500 \text{ ms}^{-1}$$

$$= 1800 \text{ km h}^{-1}$$

30. (1)

Sol. Due to surface tension, B will keep its area as minimum as possible.



So, the thread will become concave on seeing from side A.

31. (1)

Sol. Since the excess pressure due to surface tension is inversely proportional to its radius, it follows that smaller the bubble, greater is the excess pressure. Thus, when the larger and the smaller bubbles are put in communication air starts passing from the smaller into the large bubble because excess pressure inside the former is greater than inside the latter. As a result, the smaller bubble shrinks and the larger one swells.

32. (2)

Sol. $P_A + \frac{1}{2}\rho v_A^2 = P_B + \frac{1}{2}\rho v_B^2$

$$P_A - P_B = \frac{1}{2}\rho(v_B^2 - v_A^2)$$

$$0.02 \times 12000 \times 10 = \frac{1}{2} \times 1000 (v_B^2 - 20.2)$$

$$4.8 = v_B^2 - 20.2 \Rightarrow v_B = 5 \text{ m/s}$$

$$\Rightarrow A_3 v_B = A_1 v_1 + A_2 v_2$$

$$\Rightarrow 30 = 4v_1 + 8$$

$$\Rightarrow 4v_1 = 22 \Rightarrow v_1 = \frac{22}{4} = 5.5 \text{ m/s}$$

33. (3)

Sol. $\therefore$ Upward lift = pressure difference $\times$ area of wing [$\because F = p \times A$]

From Bernoulli's theorem,

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$$

Pressure difference is

$$p_2 - p_1 = \frac{1}{2}\rho(v_1^2 - v_2^2)$$

$$\text{upward lift} = (p_2 - p_1) \times A = \frac{1}{2}\rho(v_1^2 - v_2^2)A$$

34. (1)

Sol. Clothes containing oil or grease stains cannot be cleaned by water wash as the angle of contact between the oil/grease and water is obtuse.

35. (3)

Sol. Young modulus $Y = \frac{W}{A\Delta l}$

l, A and Δl are the same for both the wires.

$$Y \propto W$$

$$\frac{Y_s}{Y_b} = \frac{W_s}{W_b} = \frac{2}{1}$$

$$W_s : W_b = 2 : 1$$

36. (1)

$F = \text{rate of change of momentum}$

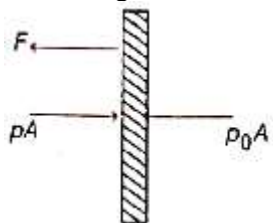
$$= \frac{\Delta p}{\Delta t} = (nm)(2v \cos \theta)$$

$$\text{Pressure, } p = \frac{F}{A} = \frac{(nm)(2v \cos \theta)}{A}$$

$$= \frac{10^{23} \times 3.32 \times 10^{-27} \times 2 \times 1000 \times \cos 45^\circ}{2 \times 10^{-4}}$$

$$= 2.34 \times 10^3 \text{ Nm}^{-2} \text{ or } 2.34 \times 10^3 \text{ Pa}$$

37. (2) Volume of the gas V is constant.



$\therefore p \propto T$, i.e. pressure will be doubled when temperature is doubled.

$$\therefore p = 2p_0$$

Now, let F be the tension in the wire. Then, equilibrium of any one piston will give

$$F = (p - p_0)A \\ = (2p_0 - p_0)A = p_0A$$

38. (1)

$PV = RT$ for 1 mole

$$W = \int PdV = \int \frac{RT}{V} dV$$

$$V = CT^{2/3}$$

$$\therefore dV = \frac{2}{3} CT^{-1/3} dT \text{ or } \frac{dV}{V} = \frac{2}{3} \frac{dT}{T}$$

$$\therefore W = \int_{T_1}^{T_2} RT \left[\frac{2}{3} \right] \frac{dT}{T} = \frac{2}{3} R(T_2 - T_1) = 166.2$$

39. (2)

AB isochoric process, $\Delta W = 0$

BC isobaric expansion, $\Delta W = (+)$ ve

CD isothermal process as

$$T_C = \frac{P_0 \cdot 3V_0}{nR} = T_D = \frac{\frac{3}{2} P_0 \cdot 2V_0}{nR}, \text{ so } \Delta U = 0$$

DA isobaric compression, $\Delta W = (-)$ ve

40. (3)

In adiabatic compression $\Delta Q = 0$ and

$\Delta W = (-)$ ve $\Rightarrow \Delta U = (+)$ ve

$\Rightarrow \Delta T = (+)$ ve, i.e., temperature increases. So, Reason is false.

41. (4)

For a diatomic gas, $C_V = \frac{5}{2} R$ and $C_P = \frac{7}{2} R$.

The work done in an isobaric process,

$$W = P(V_2 - V_1) = nRT_2 - nRT_1 \text{ or } T_2 - T_1 = \frac{W}{nR}$$

The heat given in an isobaric process is

$$Q = nC_P(T_2 - T_1) \\ = nC_P \frac{W}{nR} = \frac{7}{2} W$$

$$= \frac{7}{2} \times 2000 \text{ J} = 7000 \text{ J} = 7 \text{ xJ}$$

42. (3)

For isothermal expansion, temperature remains constant.

$$\Rightarrow PV = \text{constant}$$

$$\text{i.e., } PV = P' \times 2V \quad [\because V = 2V]$$

$$P' = \frac{P}{2}$$

For adiabatic expansion, $PV^\gamma = \text{constant}$

$$\Rightarrow \frac{P}{2} [2V]^{5/3} = P'' [16V]^{5/3}$$

$$\frac{P}{2} [2V]^{5/3} = P'' [16V]^{5/3} \quad \left[\because \gamma = \frac{5}{3} \right]$$

$$P'' = \frac{P}{2} \left[\frac{2V}{16V} \right]^{5/3} = \frac{P}{2} \left[\frac{1}{8} \right]^{5/3}$$

$$P'' = \frac{P}{64}$$

43. (4)

A-(I); B-(II); C-(IV); D-(III)

44. (2)

According first law of thermodynamics

$$dU = dQ - dW$$

$$dU = 8 \times 10^5 - 6.5 \times 10^5$$

$$= 1.5 \times 10^5 \text{ J}$$

In the second process, dU remains the same.

$$\therefore dW = dQ - dU$$

$$= 1 \times 10^5 - 1.5 \times 10^5$$

$$\therefore \text{Work done on the gas is } 0.5 \times 10^5 \text{ J}$$

45. (2)

As, efficiency is given as, $\eta = 1 - \frac{T_2}{T_1}$,

So that efficiency depends upon both the temperature of heat reservoir and sink and η is maximum when $\eta = 1$, which is when $T_2 = 0$.

CHEMISTRY

46. (4)

Sol: The elements of the group 14 form the covalent hydrides of type MH_4 . The number of the hydrides and their thermal stability also the ease of their formation decreases as we move from top to bottom in the group.

From CH_4 to PbH_4 the reducing power increases. The carbon forms the large number of the cyclic and the acyclic hydrides known as the hydrocarbons. Similarly Si and Ge also form hydrides of the general formula M_nH_{2n+2} .

The thermal stability decreases as we go down because of the increases in the size of metal.

Thus bond length increases and because of that the bond strength decreases and it results decreases in thermal stability.

47. (2)

Sol: Ga^{+1} acts as reducing agent because Ga^{+3} is more stable than Ga^{+1}

48. (2)

Sol: B give B_2O_3 & BN.

49. (3)

Sol: Both statements are correct

50. (4)

Sol: PbI_4 does not exist as Pb^{+4} is unstable due to inert pair effect.

51. (1)

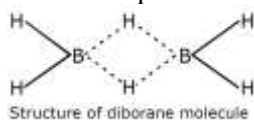
Sol: $Na_2[B_4O_5(OH)_4] \cdot 8H_2O$

52. (4)

Sol: Double salts like metal double sulphate of aluminium is Alum. Copper sulphate with aluminium sulphate salt is not alum and it is toxic in nature.

53. (3)

Sol: Each boron atom in diborane (B_2H_6) is sp^3 hybridised. In the structure of diborane four H-atoms, two on the left and two on the right, known as terminal hydrogens, are in different environments from the other two hydrogen atoms which are known as bridging atoms. The two boron atoms and the four terminal hydrogen atoms lie in the same plane while the two boron atoms and the two bridging hydrogen atoms, one above and the other below, lie in a plane perpendicular to this plane.



54. (1)

Sol: NH_3^{2+} and CH_3^+ are planar.

55. (3)

Sol: $[NF_3, H_3O^+]$

$\rightarrow sp^3$

$[NO_3^-, BF_3]$

$\rightarrow sp^2$

56. (3)

Sol: Br_2O , Cl_2O , OF_2 is the order of decreasing bond angle. It is due to the lone pair.

57. (3)

Sol.

58. (1)

Sol: Boiling point, pH & emf of a cell are intensive properties.

59. (2)

Sol: $2CO_{(g)} + O_{2(g)} \rightarrow 2CO_{2(g)}$

$\Delta n_g = 2 - 3 = -1$

$\Delta H = \Delta U + \Delta n_g RT$

$\Delta H = \Delta U - RT$

$\Delta U = \Delta H + RT$

$\Delta U > \Delta H$

Or

$\Delta E > \Delta H$

60. (1)

Sol: $U_2 - U_1 = q + w$

$U_2 - U_1 = -450 + 600$

$U_2 = 150 + E_1$

61. (3)

Sol.

62. (1)

Sol: $= \Delta H_{\text{comb}}(\text{Cyclohexene}) + \Delta H_{\text{comb}}(H_2) - \Delta H_{\text{comb}}(\text{Cyclohexene})$
 $= -3800 + (-241) - (-3920)$
 $= -4041 + (3920)$
 $= -121 \text{ KJmol}^{-1}$

63. (3)

Sol: For adiabatic process $\therefore q = 0$
 According to FLOT $\therefore \Delta E = q + W$
 $\Delta E = W$
 So work is independent of path.

64. (4)

Sol: At equilibrium $\Delta G = 0$

$\therefore \Delta G = \Delta H - T\Delta S$

$0 = \Delta H - T\Delta S$

$\Delta H = T\Delta S$

65. (1)

Sol.

66. (3)

Sol. $\Delta n = 1 - 0 = 1$ $\Delta H = \Delta E + \Delta nRT$

$$-28 = \Delta E + \frac{1 \times 2 \times 300}{1000}$$

$$-28 = \Delta E + 0.6$$

$$\Delta E = -28.6 \text{ kcal}$$

67. (3)

Sol. $\text{H}_2\text{O}(l) \rightarrow \text{H}_2\text{O}(g)$

$\Delta S = +ve$

68. (1)

Sol. $\Delta S = \frac{\Delta H_v}{T} = \frac{186.5}{373} = 0.5 \text{ kJ/K}$

69. (3)

Sol. Change in entropy is zero.

70. (4)

Sol. $\Delta G = \Delta H - T\Delta S$

If $\Delta H = +ve$, $\Delta S = -ve$

$\Delta G = +ve$ means non-spontaneous in nature

71. (4)

Sol. $T = \frac{\Delta H}{\Delta S} = \frac{170 \times 1000}{170} = 1000 \text{ K}$

$T > 1000 \text{ K} \Rightarrow$ spontaneous reaction

72. (4)

Sol. Randomness decrease while liquid crystallizes to solid. Hence, entropy will decrease.

73. (1)

Sol. For endothermic reaction

$\Delta H = +ve$ $\Delta S = (-)$

$\Delta G = \Delta H - T\Delta S$

$\Delta G = (+)$; reaction is not possible at any temperature

74. (4)

Sol.

75. (1)

Sol.

76. (3)

Sol.

77. (2)

Sol.

78. (3)

Sol.

79. (2)

Sol.

80. (3)

Sol.

81. (4)

Sol.

82. (1)

Sol.

83. (3)

Sol.

84. (4)

Sol.

85. (3)

Sol.

86. (1)

Sol.

87. (1)

Sol.

88. (3)

Sol.

89. (3)

Sol.

90. (4)

Sol.